

Synthium: Mushrooms to facilitate the transition to the Post-Anthropocene

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Figure 1: Design Iterations of *Synthium*, from left to right: A. Conceptualisation, B. Low-fidelity, C. Mid-fidelity, D. Hi-fidelity V1, E. Hi-fidelity V2

ABSTRACT

The United Nations has developed a set of Sustainable Development Goals (SDGs) as a call to action to promote prosperity while protecting the planet [16]. SDG 11 aims to make cities inclusive, safe, resilient, and sustainable, but there is a challenge to encourage adoption of a post-anthropocentric mindset that considers all stakeholders equally. We present *Synthium*, an interactive, tangible installation design that allows users to interact with other living organisms in a ludic way. We used an iterative design process to develop a multisensory system utilizing mushroom frequencies to enhance communal green spaces (see Figure 1). This prototype encourages restoration of humans' relationship with nature within urban environments: the first step in achieving a post-anthropocentric mindset leading to sustainable city development.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

Bio-Design, Sustainability, Smart Cities, Sustainable Development Goals

ACM Reference Format:

Aidan Puse, Lillian Knothe, Brooklyn Tanovic, and Shiraz Center. 2023. *Synthium: Mushrooms to facilitate the transition to the Post-Anthropocene*. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3583842>

1 INTRODUCTION

The United Nations Sustainable Development Goal 11: Sustainable Cities and Communities (SDG 11) aims to incorporate inclusiveness, safety, resilience and sustainability into city development [16]. Smart Cities, initially defined as a city that monitors and integrates conditions of all of its critical infrastructures [7], were considered to have potential in achieving this SDG [8]. However, previous literature showed that citizens, industries, and governments who build Smart Cities must first evolve their mindsets to include a post-anthropocentric outlook [17] considering all stakeholders equally, ensuring human needs are equally balanced with those of all other stakeholders, including non-humans. To develop a solution to this problem, we closely followed the Double Diamond Framework of divergent and convergent thinking shown in Figure 2. Extensive primary research found a need for widespread adoption of encompassing and comprehensive stakeholder consideration. The general public's connection to nature seemed inextricably linked to the

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3583842>

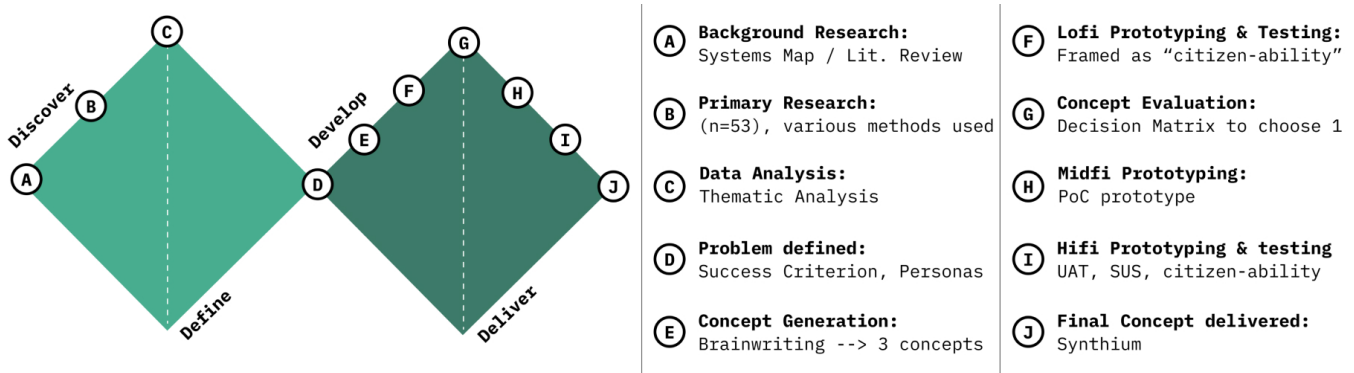


Figure 2: Design process & methods used throughout the project

incorporation of this mindset. Pivoting the project to enhance the interactivity and playfulness of communal green spaces enabled the revitalisation of laypeople's affinity to the natural world. Through divergent concept generation using design artefacts manifested directly from research findings, and appropriate convergent low-, mid-, and high-fidelity prototyping, iterated with "Citizen-Ability" [5] testing, our design solution was created: *Synthium*, which adds to the research of tangible interfaces with living displays [3] [9] and shared visions of future cities [13]. A tangible manifestation of successful collaboration between human and non-human entities, the purposeful and considered design of *Synthium* facilitates the restoration of the human relationship with the natural within urban environments. This restoration assists in the public's transition to a post-anthropocentric mindset, leading to truly sustainable smart city development by enabling a collaborative & ludic experience.

2 DISCOVERY

2.1 Background Research

Initial investigation into the problem space began with background and secondary research. Using a scholarly search engine with various key words such as "sustainable" and "city", we found the concept of Smart Cities to be closely related to the problem of Sustainable Cities and Communities. It was discovered that sustainability is a common attribute of Smart City proposals [11] [10] and that Smart Cities have "the full potential" to achieve the SDG of Sustainable Cities and Communities [8]. Subsequently, we decided to narrow our focus of Smart Cities research, resulting in two key research questions: *What are the characteristics of sustainable smart city development, and how important is the role of human behaviour in this?* These research questions were investigated through a review of existing Smart City literature represented visually in a Systems Map. Based on themes identified in Yigitcanlar et al.'s Literature review [17], we uncovered the concept of a **Post-Anthropocentric** approach to developing Smart & Sustainable Cities. The themes Yigitcanlar et al. identified centred on moving away from the notion that urban spaces are separate from nature and designed for only humans, instead moving toward embracing a sustainable, more-than-human perspective that considers new ways to cater for our broader ecological entanglements [17], including through a circular economy, to achieve sustainable Smart Cities [1]. This systems mapping

exercise also uncovered previous research into design approaches related to designing for Smart Cities and Sustainable Communities [6]. This research informed the creation of a discussion guide utilised to conduct Subject Matter Expert (SME) semi-structured interviews with members of the Smart Urbanism Lab at The University of Sydney. The purpose of the interviews was to have an open-ended discussion about the workshop and its outcomes, as well as the themes found in Yigitcanlar et al.'s Literature review, with an emphasis on conferring the Post-Anthropocentric mindset. Members collectively agreed on the literature review's findings and emphasised the need for a collaborative, post-anthropocentric mindset across all human stakeholders in order to build truly sustainable Smart Cities. There was further acknowledgement of a gap engendering such a mindset in citizens.

With this background research to inform us, we performed a re-framing exercise as defined in [14, p.128-129] to considering different perspectives, resulting in an unlocking of a broader range of solutions. The exercise shifted focus to the intrinsic connection between human and nature/natural elements, which would be inextricably linked to the post-Anthropocene. We redefined the problem space to include SDG sub-goals 11.3 and 11.7, and created an overarching design goal to solve with the design solution: **To further connect humans to the natural world within urban environments to assist in the transition to a post-anthropocentric approach to Smart and Sustainable City development.**

2.2 Primary Research

We approached primary research with a goal of understanding the workings of the current connection existing between humans in cities and the natural world. To do this, we approached research with a query: *What is holding people back from engaging in sustainable practices in their everyday lives?* The research question was re-framed from a post-anthropocentric mindset toward sustainable practices, as we intended to create User Personas to communicate user needs. Understanding these needs would allow effective generation of concepts that addressed the design goal.

We identified the primary research participants using Stakeholder Maps and Context diagrams informed by previous research: Academics, Citizens, and Non-human stakeholders were the most relevant stakeholders. Citizens would be the main focus of primary

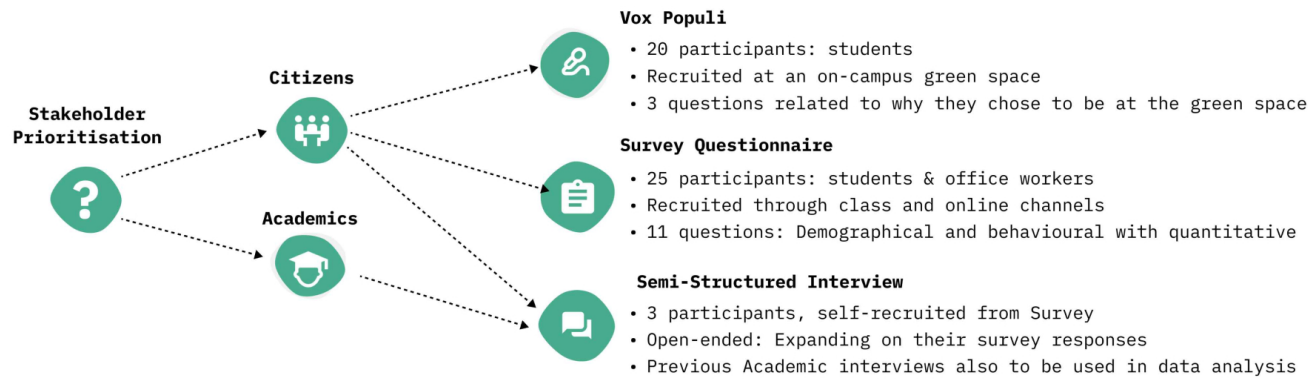


Figure 3: Primary research methods used

research, with previous SMEs being incorporated in the data analysis as well. An emphasis was placed on university students aged 18-24 as participants as the Stakeholder Maps conceptualised this demographic as active citizens of future cities, and were recruited as such. Non-human stakeholders would be addressed later in the design process. We used research methods shown in Figure 3 to collect data. Vox Populi is a research method that involves conducting short informal interviews with random members of the public. In this study, they were conducted with citizens on a campus green space to gain a large number of answers on the topic in a relatively short time frame. A survey questionnaire with 11 questions was also created to acquire more quantitative data, exploring the value and meaning of connection to nature from the perspective of citizens, including office workers to uncover the nature that exists for “current” citizens. The survey also collated key demographic data for use in Persona creation. Additionally, semi-structured interviews were conducted to allow survey participants to expand on the points that they had recorded within their survey responses, in order to gather more qualitative data, such as intrinsic motivations and participant anecdotal experiences.

3 DATA SYNTHESIS & RESULTS

To synthesise the primary research data we employed the use of a Thematic Analysis, a qualitative method for identifying, analyzing, organizing, describing, and reporting themes found within a data set. This type of analysis can produce trustworthy and insightful findings when performed rigorously [2] and would later drive the creation of important design artefacts. In summary, 279 raw data points were sourced from individual surveys, Vox Populi responses, and observational notes derived from analysis of interview transcripts consisting of both citizens and academics. We initially extracted 41 first-level themes and synthesised them into 9 second level themes, and finally organised into three third-level themes, which were as follows:

- (1) **Citizens want to spend time in green spaces.** Many Citizens have a connection to nature and value this connection. Citizens are also aware of nature’s restorative benefits, with many participants noting that their well-being is improved with and through it. However, there is a failure to prioritise

this connection, often due to personal factors eliciting fewer visits to green spaces, despite a desire to visit them.

- (2) **The need for a more-than-human design approach.** Proper qualitative analysis supported the findings from SME interviews, uncovering important requirements for concept generation, such as systems moving away from techno-centrism. Empirical evidence also showed that citizens had a one-way connection to nature, focusing mainly on the personal benefits of green spaces.
- (3) **Throughout the development of a Smart City, design considerations and citizen education must continuously evolve in order to maintain sustainability.** The concepts built today for future Smart Cities should focus on teaching citizens sustainable behaviours, with an essential focus on collaboration, to ensure the success and sustainability of Smart Cities through consideration of all stakeholder needs. *“Smart Citizens are fundamental to the “smartness” of a Smart City.”*

3.1 Design Artefacts

Subsequently, we re-framed the second-level themes into criterion for a successful design and created Personas as described in [14, p.122-123] which would communicate user needs. Using these design artefacts allowed us to generate concepts that would be successful in working toward the design goal. In Summary, the criterion found that the solution must be a physical, interactive product that aims to create a multi-sensory experience by stimulating at least two of the user’s senses, and directly connect humans to non-human stakeholders (such as flora and/or fauna) with a goal of encouraging prioritisation of this connection. It must give agency to all involved non-human stakeholders, be applicable in local and wider contexts, facilitate a movement from Anthropocene to Post-Anthropocene, and utilize the multifaceted connections amongst and between all inhabitants of a smart city environment to ensure their sustained well-being. Additionally, it should facilitate a connection to nature despite modern day factors that inhibit it and utilize the therapeutic and restorative properties of nature to improve overall well-being of humans and non-humans.



Figure 4: Musical Mushrooms; made with a speaker hidden under a cardboard box, paperclips and clay mushrooms

4 DEVELOPING CONCEPTS

To generate concepts, we used the “Brainwriting 6-3-5” method described in [14, p.32-33], ensuring each concept was informed by the defined Design Artefacts as well as additional Smart City concepts, such as the “5 A’s Framework” [12]. 12 concepts were generated and subsequently narrowed down to the 3 most effective, shown in Figure 4 - 6. **Concept 1: Musical Mushrooms** (Figure 4) is interactive light, sound, and touch system that utilizes the frequencies naturally emitted and conducted by mushrooms to highlight the natural processes of the environment. **Concept 2: Personified Plant** (Figure 5) is a plant featuring a hologram that responds to physical touch and audio response, giving plants human-like features to increase human’s empathy for them. **Concept 3: Breathing Tree** (Figure 6) is an integrated LED large plant installation that visualizes and audiates a tree’s natural ability to capture carbon dioxide and release oxygen, allowing users to view flora as more sentient. These were then transformed into low-fidelity prototypes made from materials on hand, allowing us to gather early user feedback and select the best concept for development into a high-fidelity prototype. The interactions were facilitated with the “Wizard of Oz” method, controlled manually by a team member.

4.1 Design Evaluation

Testing the concepts was performed on three facets as shown in Figure 7, and most importantly, re-framed toward “Citizen-ability” testing, rethinking the user as part of a much larger, “more-than-human”, more complex ecosystem of cohabitation [5], decentralising the human in the design of collaborative cities [4]. This ensured a responsible design process by considering the more-than-human factors in the explored solutions by exercising more-than-human thinking when evaluating them. To do this, User Journey Maps (UJMs) were made for each concept, a design tool which describes not only describes what a user may be doing and feeling, but also describes the supporting infrastructure [14, p.162-163], which enabled a consideration of the broader effects of each of the interaction touch-points. The prototypes were tested with 5 users each. Observations following testing were synthesised into a “Strengths,

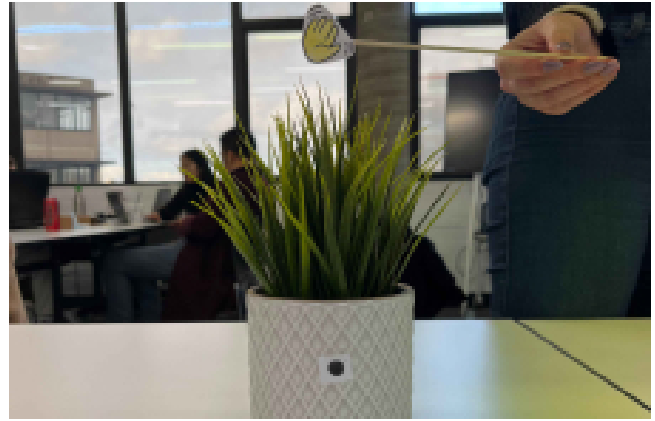


Figure 5: Personified Plant; made with hand drawn emoticons on skewers, a potted plant, and a hand-drawn button



Figure 6: Breathing Tree; made with a small plastic Christmas tree, clay sensors, and coloured strips of paper

Weaknesses, Opportunities” (SWO) table for each concept and discussions were centred around the “citizen-ability” perspective of each concept by reviewing the SWO table and the UJMs. These helped inform a decision matrix, used to evaluate the success of each prototype and choose the final solution, eliminating personal bias by quantitatively assessing the benefits and consequences of each concept as described in [14, p.54-55]. The matrix considered the problem space, criteria for success, and feasibility concerns, giving some variables more importance than others based on the context of the design goal. Results revealed that “Musical Mushrooms”, renamed to *Synthium*, was the most appropriate concept in context of the design goal, and therefore was chosen to be further developed, which also involved creating a mid-fidelity proof-of-concept prototype (Figure 1-C) before moving onto high-fidelity prototyping.

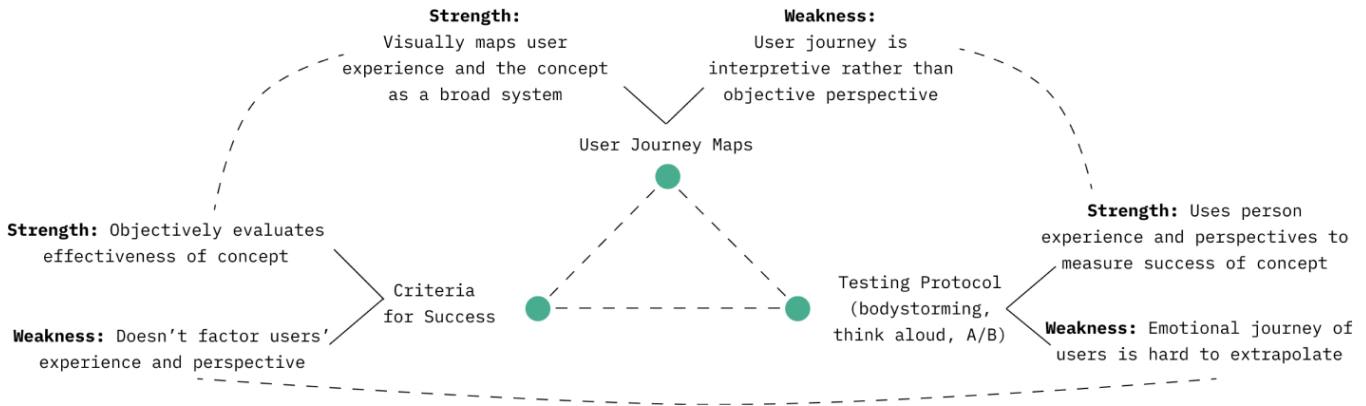


Figure 7: Citizen-ability testing methodology employed during the Low-Fidelity testing phase.

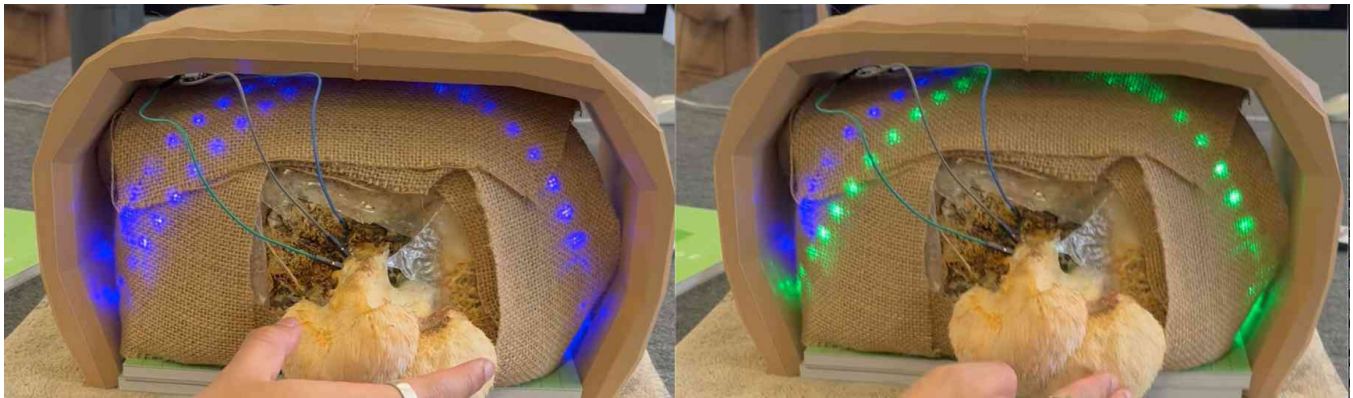


Figure 8: Synthium exhibiting its functionality. A: sounds & light produced from mushrooms B: sounds & light produced after user's touch is detected

5 FINAL CONCEPT: SYNTHIUM

Synthium is an interactive light and sound system, which utilises the naturally occurring frequencies produced by mushrooms. The frequency read is converted into pitch and expressed audibly through a synthesiser. This is referred to as “bio-data sonification”, producing ambient, low fidelity sounds in response to user’s touch with the mushrooms. A strip of LEDs also visualises the audio expressed, creating a multi-sensory experience that illuminates the natural processes of the environment. Once mushrooms approach maturation, harvesting for human consumption is encouraged. Disposing of sporing Mushrooms & expended *Synthium* “Pod” substrate through composting is also possible and encouraged. These disposal methods enable active Smart Citizen engagement within a circular economy.

A single *Synthium* pod (Figure 8) is a block of substrate with an active Mycelium culture which fruits into mushrooms. The block is wrapped with hessian fabric embedded with an LED strip, covered by a 3D printed housing over the pod. The embedded LED strip and two sensors plugged into the mushroom connect to an Arduino, which enable two kinds of functionality: one from the mushrooms themselves, and the other via user interaction with the system. For Mushrooms, a Galvanic Skin Response (GSR) sensor detects

changes in the electrical conductivity of mushrooms and converts this into MIDI packets (notes) that are sent to music software on a connected computer, which then plays the corresponding sound through a speaker. Additionally, a flash of blue light is emitted through the embedded LED strip shown in Figure 8-A. For users, a Capacitor sensor detects touch on the mushrooms and converts it into MIDI packets containing chords, which are sent to music software on a connected computer, resulting in a corresponding sound played through a speaker and green light traveling through the LED strip (shown in Figure 8-B). The notes defined by the mushroom functionality directly determines the chords played in response to touch. Humans and non-human stakeholders were considered equally by creating non-human personas [15] to represent their needs as shown in Figure 9, in coalition with Bio Design expert Dr. Phillip Gough. Some key needs found from this exercise included a constant high humidity and frequent water misting, which was addressed by assembling the final design in a custom greenhouse with an autonomous water mister, found in Figure 10. To further the experience of *Synthium*, the greenhouse was also lined with LED lights that “pulsated” when users touched the mushrooms. Multiple pods contained in a single greenhouse coalesced into an interactive

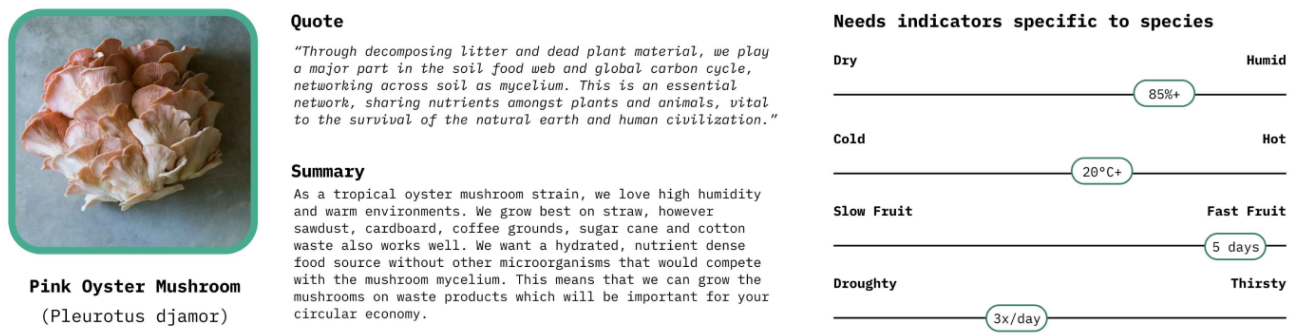


Figure 9: Example of a non-human persona created: Pink Oyster Mushroom

and ludic experience where Mycelium and humans collaborated to create music, building empathy between one another.

5.1 Final Evaluation

To evaluate the success of the final concept in meeting the stakeholder needs and the design goal, we performed one more round of “citizen-ability” testing framing the user as city resident [5](an example in Figure 10). Testing used User Acceptance Testing (UAT) methods and open-ended scenarios, concluding with a Systems Usability Scale (SUS) Test, as described in [14, p.160-161]. Participants were presented with a scenario based in a near-distant future smart city and were tasked with using the concept in context with the scenario given, with the SUS test then providing participant evaluation on whether or not the system was effective at meeting the user needs based on the scenario. Findings from five participants revealed an average score of 86, indicating the system was usable and effective at meeting the user needs. Participants also noted that it was effective at increasing their connection to nature through the lense of reducing “Mycophobia”, with participants initially hesitant to interact with the prototype, however after the test concluded viewing mushrooms more positively and by extension more equally.

6 CONCLUSION & FUTURE WORK

In response to SDG 11: Sustainable Cities and Communities and its subgoals 11.3 & 11.7, we conceptualised, iterated and developed a solution with a design goal to further connect humans to the natural world within urban environments, for the express purpose of assisting in the transition to a post-anthropocentric approach to Smart and Sustainable City development. The final result was *Synthium* - an interactive light and sound system, which utilises the naturally occurring frequencies produced by mushrooms. Multiple rounds of “citizen-ability” testing revealed *Synthium* was able to effectively meet the needs of users and the problem space and was reaffirmed with SUS test. *Synthium* was conceptualised directly from findings of extensive secondary and primary research. Furthermore, *Synthium* was subject to multiple iterations and appropriate evaluation. Therefore, we evaluate *Synthium* as a successful design solution in response in our design goal to reconnect humans to the natural world: the very first and step toward the post-anthropocentric mindset needed to solve SDG11: Sustainable Cities & Communities.

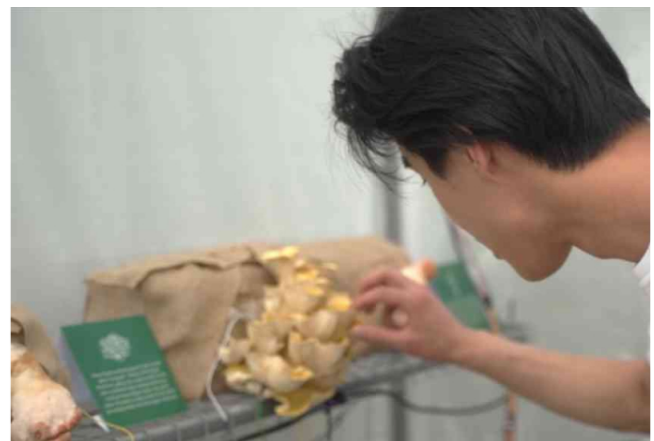


Figure 10: Participant interacting with Synthium in the greenhouse during testing

Synthium is a complete and functional high-fidelity prototype. Beyond its current successful iteration we have identified future work in this problem space that could further develop the potential of the *Synthium* system. Speculation with our non-human stakeholder coalition conceptualised the possibility of spore collection technology in a future smart city, which would allow for responsible dispersal of *Synthium* system spores. Additionally, we have identified more sustainable options for housing such as using grown and baked mycelium. We also see potential in researching Mycophobia and understanding the psychology behind this phobia, in order to develop methods of resolution that could make the system more accessible to a wider range of users to promote a greater connection to nature for all. Finally, we also encourage our fellow students, designers, and researchers to embark on further work into solutions to instill a post-anthropocentric mindset from a similar or different frame to ensure SDG11 is eventually met.

ACKNOWLEDGMENTS

We would like to extend our thanks to the students and citizens who participated in our Primary Research on the University of Sydney Campus and online, as well as the members of the Smart Urbanism Lab who helped us understand the Problem Space. Thank

you to Dr. Phillip Gough who acted in our coalition for the Non-Human Stakeholders, and Dr. Callum Parker, who was our Unit Coordinator for the class in which this work was conducted as a part of our University coursework, and finally, thanks to Youtuber “MycoLyco,” for introducing the concept of “bio-data-sonification” that was needed for the concept after an email enquiry. We would also like to thank our kind reviewers for the feedback and for the additional references which we have included.

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